Topic 5 — Blood, Heart, Circulation and Lymph

Sources baked in: NCERT Class 11 (Body Fluids and Circulation), Landsteiner (1900), Einthoven ECG, Lucent General Science (Circulatory System), UPPCS Prelims PYQs (2018–2025) and UKPCS Prelims **Weight:** ★★★ High — heavily tested: number of heart chambers, pulmonary vein/artery oxygenation exception, blood group compatibility, blue baby methemoglobinemia, universal donor/recipient, and heart sound valve mechanics **Last verified:** September 2026

Current Affairs

Year	Event / Fact	Must-Score Fact
Historical & Still Tested	First human heart transplant performed on **3 December 1967** by **Dr. Christiaan Barnard** at Groote Schuur Hospital, Cape Town, South Africa.	Frequently tested in state PSCs; William Harvey discovered blood circulation (1628), but did **not** perform heart transplants.
National Landmark	First successful human heart transplant in India performed on **3 August 1994** by **Dr. P. Venugopal** at **AIIMS, New Delhi**.	Landmark in Indian cardiothoracic surgery.
Still Current	**World Heart Day** is observed annually on **29 September**; **World Blood Donor Day** is observed on **14 June** (birth anniversary of Karl Landsteiner).	Commemorates discovery of the ABO blood group system by Landsteiner.

5.1 Cellular Elements of Blood & Haemostasis

- **The Liquid Connective Matrix (Plasma)**:
- Straw-coloured, viscous fluid constituting ~55% of total blood volume.
- Plasma proteins play fundamental roles in homeostasis:
  - **Serum Albumin (~60% of total plasma protein, 3.5–5.0 g/dL)**: Synthesised exclusively by the liver; provides 80% of total **colloid osmotic pressure (oncotic pressure ~ 25 mm Hg)** that holds water inside the vascular lumen, preventing plasma from extravasating into interstitial spaces. In liver cirrhosis or severe protein malnutrition, albumin synthesis collapses, causing widespread fluid retention (**Ascites and Oedema**).
  - **Serum Globulins (~36% of plasma protein)**: α and β globulins act as transport carrier proteins (e.g. transferrin carries iron; ceruloplasmin carries copper); **γ-globulins (Immunoglobulins: IgG, IgM, IgA, IgD, IgE)** are synthesised by plasma B-cells and act as specific circulating defensive antibodies.
  - **Fibrinogen (~4% of plasma protein)**: Soluble high-molecular-weight clotting precursor synthesised by the liver.
- **Erythrocyte Morphometrics & Haemoglobin Chemistry**:
- Mature human erythrocytes are non-nucleated biconcave discs (7.2 microns diameter, 2.2 microns thick); biconcave geometry increases surface-area-to-volume ratio by 30%, maximizing gas diffusion efficiency and allowing flexibility to deform through narrow 5-micron capillary lumens.
- **Erythropoiesis**: In early embryos, RBCs are formed in the **yolk sac**; from the 2nd to 7th month of gestation, formation shifts to the **liver and spleen**; from the 7th month onward and throughout adult life, formation is restricted exclusively to **red bone marrow** (in vertebrae, ribs, sternum, pelvis, and proximal ends of femur/humerus).
- Normal destruction of RBCs: Aged, fragile RBCs are engulfed by reticuloendothelial macrophages of the spleen and liver; the globin protein is degraded into free amino acids; iron is salvaged and stored in the liver as **Ferritin**; the iron-free porphyrin ring is degraded into yellow-green bile pigments: **Biliverdin** $\rightarrow$ **Bilirubin**.
- **Leukocyte Differential Count & Diagnostic Signatures**:
- **Neutrophils (60–65%)**: Nucleus has 3–5 interconnected lobes; cytoplasm has fine neutral-staining granules; motile via amoeboid diapedesis; active phagocytes.
- **Eosinophils (2–3%)**: Bilobed spectacle-shaped nucleus; coarse spherical granules that stain bright pinkish-red with acidic dyes (Eosin); release histaminase to dampen allergic reactions; attack parasitic worms by secreting cytotoxic major basic

protein (MBP).

- **Basophils (0.5–1%)**: S-shaped nucleus; coarse irregular granules that stain intense dark blue with basic dyes (Methylene Blue); liberate **Histamine** during acute hypersensitivity (Type I anaphylaxis).
- **Monocytes (6–8%)**: Largest leukocyte ($15–20$ microns); kidney-shaped or indented nucleus; circulate in blood for 24 hours before entering connective tissues to become **Macrophages** (e.g. Kupffer cells in liver, Alveolar macrophages in lungs, Microglia in brain, Osteoclasts in bone).
- **Lymphocytes (20–25%)**: Small ($7–10$ microns); spherical, intensely staining nucleus surrounded by a thin rim of basophilic cytoplasm:
  - **B-Lymphocytes**: Mature in bone marrow; differentiate into antibody-secreting **Plasma Cells** upon antigen stimulation (Humoral Immunity).
  - **T-Lymphocytes**: Mature in the **Thymus** gland; differentiate into **Cytotoxic T-cells ($CD8^+$)** which destroy virus-infected and cancer cells, and **Helper T-cells ($CD4^+$)** which orchestrate the entire immune response (Cell-Mediated Immunity).
- **Thrombocytes & Haemostasis (The Clotting Cascade)**:
- Platelets adhere to exposed sub-endothelial collagen via von Willebrand factor, activate, and release ADP and Thromboxane A2 to form an initial **Primary Platelet Plug**.
- The intrinsic and extrinsic coagulation pathways converge on the common pathway:

$$\text{Factor X} \xrightarrow[\text{Thrombokinase}]{\text{Ca}^{2+}} \text{Active Factor Xa}$$

$$\text{Prothrombin (Factor II)} \xrightarrow[\text{Ca}^{2+}]{\text{Factor Xa + Va}} \textbf{Thrombin (Factor IIa)}$$

$$\text{Soluble Fibrinogen (Factor I)} \xrightarrow{\text{Thrombin}} \textbf{Insoluble Fibrin Monomers} \xrightarrow{\text{Factor XIIIa}} \textbf{Stable Cross-Linked Fibrin Clot}$$

- Clot retraction occurs as platelets contract via thrombosthenin, squeezing out a clear pale-yellow fluid called **Serum**.
- **Specialized Comparative & Diagnostic Hematology Facts:**
- **Myoglobin Contains Iron (Fe):** Iron- and oxygen-binding chromoprotein found in striated skeletal and cardiac muscle tissues; contains an iron-porphyrin heme group. *(Compare: Chlorophyll contains Magnesium, Keratin contains Sulfur, Collagen is structural fibrous protein).*
- **Hemoglobin Dissolved in Plasma in Earthworm:** In **Earthworms (*Pheretima*)**, hemoglobin is dissolved directly in blood plasma rather than enclosed inside corpuscles (they lack erythrocytes).
- **WBC Dimensions & Ratios:** Human leukocytes have an average diameter of approximately $7 \text{ microns} = 0.007 \text{ mm}$. The normal physiological numerical ratio of Red Blood Cells to White Blood Cells is $600 : 1$.
- **Pathology of Blood Counts:**
  - **Leukemia:** Pathological, malignant proliferation of abnormal white blood cells in bone marrow and bloodstream.
  - **Thrombocytopenia:** Abnormal drop in circulating blood platelet count ($< 100,000/\mu\text{L}$), classically seen in acute **Dengue fever**.
- **Artificial Blood Research:** In 2006, researchers in **Britain (UK)** engineered plastic-based artificial blood molecules that mimic hemoglobin and can be administered safely across all blood groups.

5.2 Cardiac Anatomy, Electrophysiology & Haemodynamics

- **Anatomical Organisation of the Human Heart**:
- Weighs approximately $280–340 \text{ grams}$ in adult males and $230–280 \text{ grams}$ in females; approximately the size of an individual's clenched fist.
- Heart wall comprises three distinct tissue coats:
  1. **Epicardium (Visceral Pericardium)**: Outer thin protective serous layer.
  2. **Myocardium**: Middle thick contractile layer composed of branching **Cardiac Muscle Fibres** connected by transverse **Intercalated Discs** containing low-resistance **Gap Junctions**, allowing the entire chamber to contract as a functional electrical syncytium.
  3. **Endocardium**: Inner smooth lining of simple squamous endothelium continuous with the endothelial lining of systemic blood vessels.

- **Surgical Milestones — Heart Transplantation**:
- **World's First Human Heart Transplant:** Successfully performed on **3 December 1967** by South African cardiothoracic surgeon **Dr. Christiaan Barnard (Dr. Christian Bernard)** at Groote Schuur Hospital, Cape Town, South Africa (patient: Louis Washkansky). *(Note: William Harvey discovered the circulation of blood in 1628, but did not perform transplants).*
- **India's First Heart Transplant:** Successfully conducted on **3 August 1994** by **Dr. P. Venugopal** at the All India Institute of Medical Sciences (**AIIMS, New Delhi**).
- **Internal Partitioning & Flow Architecture**:
- The right and left atria are separated by the **Interatrial Septum**, which displays an oval shallow depression called the **Fossa Ovalis** (remnant of the embryonic *Foramen Ovale*, through which blood bypassed the unexpanded fetal lungs).
- The right and left ventricles are separated by the thick, muscular **Interventricular Septum**.
- **Blood Flow Sequence Through Chambers**:

$$\text{Venae Cavae} \rightarrow \text{Right Atrium} \xrightarrow[\text{Valve}]{\text{Tricuspid}} \text{Right Ventricle} \xrightarrow[\text{Semilunar}]{\text{Pulmonary}} \text{Pulmonary Artery} \rightarrow \text{Lungs}$$

$$\text{Lungs} \xrightarrow{\text{Pulmonary Veins}} \text{Left Atrium} \xrightarrow[\text{Mitral}]{\text{Bicuspid}} \text{Left Ventricle} \xrightarrow[\text{Semilunar}]{\text{Aortic}} \text{Aorta} \rightarrow \text{Systemic Organs}$$

- **Electrophysiology of the Cardiac Pacemaker**:
- Cardiac muscle is **myogenic**: the electrical impulse originates intrinsically within specialized auto-rhythmic nodal cardiac muscle cells without requiring nervous stimulation.
- **SA Node (Sinoatrial Node / Pacemaker)**: Located in the right atrium; initiates spontaneous rhythmic cardiac impulses at 70–75 beats/minute.
- Autonomic Regulation:
  - **Sympathetic Nervous System (Cardiac Accelerator Nerves)**: Releases **Noradrenaline (Norepinephrine)**; binds to β_1-adrenergic receptors, increasing SA node firing rate, cardiac contractility, and cardiac output.
  - **Parasympathetic Nervous System (Vagus Nerve / CN X)**: Releases **Acetylcholine (ACh)**; binds to muscarinic receptors, hyperpolarizing the SA node and **slowing the heart rate (Bradycardia)**.
- **Physical Dynamics of Blood Pressure & Measurement**:
- Normal resting adult blood pressure is standardized as $120/80\ \text{mm Hg}$ (Systolic $120\ \text{mm Hg}$ / Diastolic $80\ \text{mm Hg}$).
- **Pulse Pressure:** The difference between systolic and diastolic pressure = $120 - 80 = 40\ \text{mm Hg}$.
- Blood pressure is highest in the Ascending Aorta (~$120\ \text{mm Hg}$), declines progressively across large systemic arteries, drops sharply across high-resistance **Arterioles** (the primary site of peripheral vascular resistance), falls to $15–30\ \text{mm Hg}$ in capillaries, and drops to near zero ($0–4\ \text{mm Hg}$) in the right atrium (Central Venous Pressure / CVP).
- **Clinical Sphygmomanometry**: An inflatable cuff is wrapped around the upper arm and inflated above systolic pressure to occlude the brachial artery; as the valve is released slowly, blood begins to spurt through the partially compressed vessel, generating turbulent tapping sounds audible through a stethoscope — these are the **Sounds of Korotkoff**. The first audible tap denotes **Systolic BP**; the complete disappearance of sound denotes **Diastolic BP**.

Bilingual Terminology -- Blood, Heart and Circulation

English Term	Hindi Term	Key Anchor
Blood (रक्त)	रक्त / खून	Fluid connective tissue; plasma + RBC + WBC + Platelets
Plasma (प्लाज्मा)	रक्त प्लाज्मा	Liquid component of blood (~55%); water, proteins, salts
RBC / Erythrocyte (लाल रक्त कणिका)	लाल रक्त कोशिका	Carries O2 via haemoglobin; no nucleus in mature human RBCs
WBC / Leucocyte (श्वेत रक्त कणिका)	श्वेत रक्त कोशिका	Defence cells; 5 types (neutrophil, eosinophil, basophil, lymphocyte, monocyte)
Platelet / Thrombocyte (बिम्बाणु)	प्लेटलेट / रक्त बिम्बाणु	Blood clotting; smallest blood cells; 1.5-3.5 lakh/mm3
Haemoglobin (हीमोग्लोबिन)	हीमोग्लोबिन	Protein in RBC; iron-containing; carries O2; normal: M=14-18, F=12-16 g/dL
Blood Group (रक्त समूह)	रक्त समूह	ABO system (Landsteiner 1901); also Rh factor

English Term	Hindi Term	Key Anchor
Rh Factor (आरएच कारक)	आरएच कारक	Rh antigen on RBC; Rh+ (positive) or Rh- (negative)
Universal Donor (सार्वभौम दाता)	सार्वभौम दाता	Blood group **O-** (O Rh negative)
Universal Recipient (सार्वभौम प्राप्तकर्ता)	सार्वभौम प्राप्तकर्ता	Blood group **AB+** (AB Rh positive)
Heart (हृदय)	हृदय	Muscular pump; 4 chambers; located in thoracic cavity
Artery (धमनी)	धमनी	Carries oxygenated blood AWAY from heart (except pulmonary artery)
Vein (शिरा)	शिरा	Carries deoxygenated blood TOWARDS heart (except pulmonary vein)
Capillary (केशिका)	केशिका	Finest blood vessel; site of exchange between blood and tissues
Blood Pressure (रक्तचाप)	रक्तचाप	Systolic/Diastolic; normal = 120/80 mmHg
Cardiac Cycle (हृदय चक्र)	हृदय चक्र	One complete heartbeat; systole (contraction) + diastole (relaxation)
Lymph (लसीका)	लसीका	Colourless fluid similar to plasma; no RBCs; returns to blood
Clotting (थक्का बनना)	रक्त का थक्का जमना	Fibrin mesh traps blood cells; Vitamin K essential; platelets initiate

One-Liner Revision -- Blood, Heart and Circulation

#	Fast Fact
1	Blood = **55% plasma** + 45% formed elements (RBC, WBC, Platelets)
2	RBC lifespan = **120 days**; WBC lifespan = a few hours to several years
3	RBC count (normal) = **45-55 lakh/mm3** in men; 40-50 lakh/mm3 in women
4	WBC count (normal) = **4,000-11,000/mm3**
5	Platelet count (normal) = **1.5-3.5 lakh/mm3**
6	Most abundant WBC = **Neutrophil** (60-70%); first line of defence
7	Antibody-producing WBC = **Lymphocyte** (B-lymphocytes)
8	Haemoglobin = iron (Fe2+) + globin protein; made in **bone marrow**
9	ABO blood groups discovered = **Karl Landsteiner (1901)**; Nobel 1930
10	Universal Donor = **O-** (O negative); no A, B antigens; no Rh antigen
11	Universal Recipient = **AB+** (AB positive); has all antigens
12	Erythroblastosis foetalis = Rh incompatibility between Rh- mother and Rh+ baby
13	Heart chambers: 2 Atria (upper) + 2 Ventricles (lower) = **4 chambers**
14	Heart rate (normal resting adults) = **60-100 beats/min**
15	Heartbeat sound "Lub-Dub" = closing of **AV valves** (Lub) + semilunar valves (Dub)
16	ECG = Electrocardiogram; records electrical activity of heart
17	Blood pressure normal = **120 mmHg** (systolic) / **80 mmHg** (diastolic)
18	Hypertension = BP > 140/90 mmHg (Silent killer)
19	Arteries carry blood FROM heart; thicker walls; no valves (except at heart base)
20	Veins carry blood TO heart; thinner walls; have valves to prevent backflow
21	Pulmonary artery = ONLY artery carrying deoxygenated blood (from heart to lungs)
22	Pulmonary vein = ONLY vein carrying oxygenated blood (from lungs to heart)

#	Fast Fact
23	Coronary arteries = supply blood to heart muscle itself
24	Heart attack (Myocardial infarction) = blockage of coronary artery
25	Clotting factor = **Fibrinogen** (in plasma) converts to fibrin; **Vitamin K** needed
26	Haemophilia = genetic disorder; blood doesn't clot; X-linked recessive
27	Spleen = "Graveyard of RBCs"; also filters blood and stores platelets
28	Bone marrow = site of blood cell production (haematopoiesis)
29	Lymph = plasma that leaks from capillaries into tissue; carried in lymphatic system
30	Carotid arteries = main arteries supplying blood to the brain